Toward a South Asian-Specific Stroke and Intracerebral Hemorrhage Risk Calculator: A Two-Model Synthesis of the Available Evidence and a Map of Its Remaining Gaps

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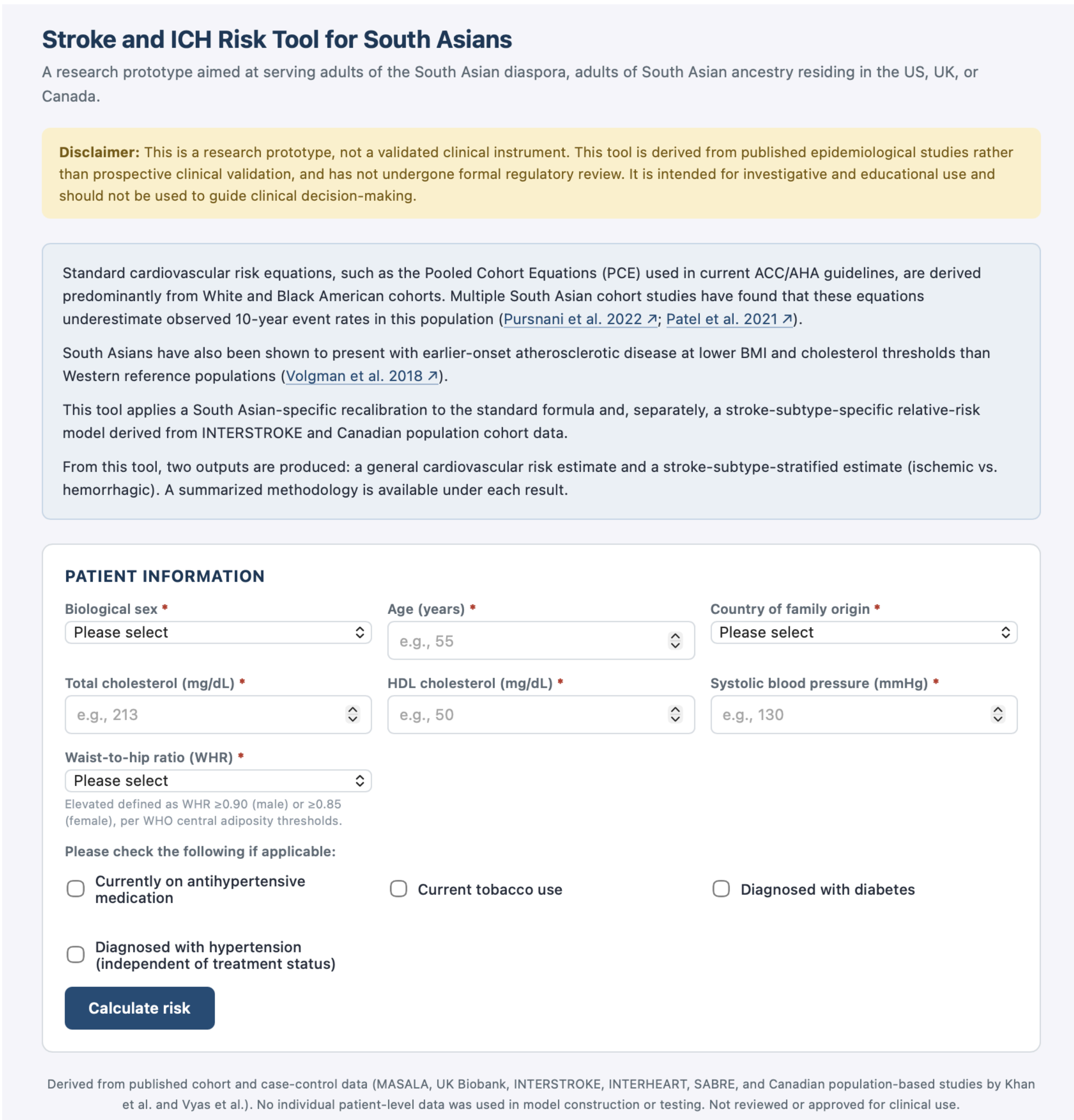

Stroke and ICH Risk Tool for South Asians

A research prototype aimed at serving adults of the South Asian diaspora, adults of South Asian ancestry residing in the US, UK, or Canada.

Disclaimer: This is a research prototype, not a validated clinical instrument. This tool is derived from published epidemiological studies rather than prospective clinical validation, and has not undergone formal regulatory review. It is intended for investigative and educational use and should not be used to guide clinical decision-making.

Standard cardiovascular risk equations, such as the Pooled Cohort Equations (PCE) used in current ACC/AHA guidelines, are derived predominantly from White and Black American cohorts. Multiple South Asian cohort studies have found that these equations underestimate observed 10-year event rates in this population (Pursnani et al. 2022 ↗; Patel et al. 2021 ↗).

South Asians have also been shown to present with earlier-onset atherosclerotic disease at lower BMI and cholesterol thresholds than Western reference populations (Volgman et al. 2018 ↗).

This tool applies a South Asian-specific recalibration to the standard formula and, separately, a stroke-subtype-specific relative-risk model derived from INTERSTROKE and Canadian population cohort data.

From this tool, two outputs are produced: a general cardiovascular risk estimate and a stroke-subtype-stratified estimate (ischemic vs. hemorrhagic). A summarized methodology is available under each result.

PATIENT INFORMATION

Biological sex *
Please select

Age (years) *
e.g., 55

Country of family origin *
Please select

Total cholesterol (mg/dL) *
e.g., 213

HDL cholesterol (mg/dL) *
e.g., 50

Systolic blood pressure (mmHg) *
e.g., 130

Waist-to-hip ratio (WHR) *
Please select
Elevated defined as WHR ≥0.90 (male) or ≥0.85 (female), per WHO central adiposity thresholds.

Please check the following if applicable:

- [ ] Currently on antihypertensive medication
- [ ] Current tobacco use
- [ ] Diagnosed with diabetes
- [ ] Diagnosed with hypertension (independent of treatment status)

Calculate risk

Derived from published cohort and case-control data (MASALA, UK Biobank, INTERSTROKE, INTERHEART, SABRE, and Canadian population-based studies by Khan et al. and Vyas et al.). No individual patient-level data was used in model construction or testing. Not reviewed or approved for clinical use.

Abstract

Background: South Asian-origin adults face substantially elevated cardiovascular and cerebrovascular disease risk that is not captured by widely used risk-prediction tools such as the American College of Cardiology/American Heart Association Pooled Cohort Equations (PCE), which were derived exclusively from White and Black cohorts. Evidence describing this excess risk is scattered across case-control studies and regional cohorts, and no published tool synthesizes this evidence into a stroke- and intracerebral hemorrhage (ICH)-specific risk estimate for South Asian-origin adults living in Western healthcare systems.

Methods: A structured synthesis of the peer-reviewed literature describing cardiovascular and cerebrovascular risk in South Asian populations was conducted. From this synthesis, we developed an evidence-synthesis–based, dual-model prototype designed to illustrate recalibration strategies for South Asian cardiovascular and stroke risk estimation. This tool is intended to demonstrate how risk estimates may differ when South Asian-specific data are incorporated. Model A recalibrates the standard PCE using a risk-dependent multiplier derived from two independent South Asian cohorts. Model B is a stroke/ICH subtype-specific multiplicative relative-risk score built from INTERSTROKE's South Asia-region odds ratios. This model is anchored to South Asian diaspora baseline incidence rates and age-scaled using stroke-subtype-specific hazard ratios from a large Canadian immigrant cohort.

Results: Internal consistency checks confirm the calculator's outputs match its calibration inputs and remain plausible across a wide range of patient profiles. Independent corroboration comes from four additional cohorts (SABRE, MASALA, INTERHEART, UK Biobank), which consistently find that standard cardiovascular risk tools underperform in South Asian populations, though the specific dimension and magnitude of underperformance varies by cohort. Two unresolved tensions in the underlying evidence are also identified: cohorts disagree on the direction of PCE miscalibration in South Asians, and on which South Asian country-of-origin subgroup carries the highest relative cerebrovascular risk.

Conclusions: This tool is presented as an open-source, freely available research prototype, not a validated clinical instrument. Four gaps in the underlying evidence base are identified: the absence of any South Asian-specific family history coefficient in the accessible literature, the absence of sex-stratified PCE calibration data for South Asians despite a targeted search, the two unresolved directional disagreements described above, and Model B's reliance on an age-stratified stroke risk curve pooled across all immigrant groups rather than South Asian-specific. Therefore, future studies should prioritize these four directions to replace the pooled-immigrant approximation currently used in Model B. As this evidence accumulates, each finding can be used to directly refine the calculator's coefficients, moving it from a synthesis-stage prototype toward a validated clinical tool.

1. Introduction

South Asian-origin individuals, comprising people with ancestry from India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, and the Maldives, represent roughly a quarter of the world's population and carry a disproportionate share of the global cardiovascular disease burden.[1] This excess risk persists in diaspora populations living in high-income Western healthcare systems, where South Asian ancestry has been formally recognized as a cardiovascular "risk-enhancing factor" in current American College of Cardiology/American Heart Association (ACC/AHA) primary prevention guidelines.[2] Despite this recognition, no South Asian-specific coefficient set exists for the ACC/AHA Pooled Cohort Equations (PCE), which is the risk calculator underlying most contemporary statin-eligibility decisions in the United States. The PCE was derived exclusively from non-Hispanic White and Black cohorts, and its own developers acknowledge it was not designed for use in other ancestry groups.[3]

The evidence needed to build a South Asian-specific alternative already somewhat exists, but it is scattered. Case-control studies such as INTERHEART and INTERSTROKE provide South Asia-region odds ratios for individual risk factors.[4,5,6] Prospective cohorts, including Pursnani et al.'s Kaiser Permanente Northern California cohort, the UK Biobank, and SABRE (Southall and Brent Revisited), provide calibration data comparing standard risk-tool predictions against observed outcomes in South Asian participants.[7,8,9] MASALA (Mediators of Atherosclerosis in South Asians Living in America), a further South Asian cohort in the same diaspora population, has independently linked elevated predicted cardiovascular risk to subclinical atherosclerosis.[16] Administrative-data studies from Ontario and British Columbia provide population-level stroke and intracerebral hemorrhage (ICH) incidence rates specific to the South Asian diaspora, along with an age-stratified, stroke-subtype-specific hazard ratio curve for immigrant populations more broadly.[10,11] No published work was identified that synthesizes these sources into a single, stroke- and ICH-specific risk tool for South Asian-origin adults.

Therefore this study therefore has two aims. The first is to build and describe such a tool: a two-model calculator that provides both a general cardiovascular risk context (recalibrated from the PCE) and a stroke/ICH subtype-specific estimate (built directly from South Asian stroke epidemiology). The second is to document, precisely, where the underlying evidence remains incomplete, so that targeted future research can address those gaps and in turn refine the calculator's coefficients, moving it toward clinical validation. Accordingly, this study presents an evidence-synthesis-based prototype designed to illustrate how recalibration strategies may alter risk estimation in South Asian populations, while explicitly mapping the limitations of current approaches.

2. Methods

2.1 Scope and Target Population

This tool is intended specifically for South Asian-origin adults living in Western, high-income healthcare systems, the same populations sampled by MASALA, the UK Biobank, and SABRE. It is not validated for people currently resident in South Asia. The diaspora and resident populations are not epidemiologically interchangeable. Khan et al. (2017) used population-based data from Ontario and British Columbia to compare stroke incidence across ethnic groups within the same Canadian health system, and found that South Asian-Canadians had substantially lower age- and sex-standardized incidence of both ischemic stroke and ICH than White Canadians in that same system.[10] This pattern is consistent with a "healthy immigrant effect," in which the diaspora population differs systematically from both the host-country population and the population remaining in the country of origin. The tool is therefore situated, deliberately, within the diaspora population: every coefficient it uses is drawn from diaspora cohorts, and its outputs should be interpreted as applying only to that population.

2.2 Literature Synthesis

An iterative, primary-source-prioritized literature synthesis was conducted, beginning with foundational case-control studies of South Asian cardiovascular risk (INTERHEART, INTERSTROKE) and expanding outward to prospective cohorts with calibration data (MASALA, UK Biobank, SABRE), administrative incidence studies (Khan et al. 2017, Vyas et al. 2021), and disaggregated electronic health record analyses (Rodriguez et al. 2019, Razieh et al. 2022).[4-13] Each source, including its verification status, sample size, confidence intervals, and interpretive notes, is logged in a structured tracking document (Supplementary File 2), an evidence ledger maintained alongside this manuscript that records where every coefficient in the calculator originates.

2.3 Model A: Recalibrated Pooled Cohort Equations

Illustrative profile. Sections 2.3 through 3.1 use a single hypothetical patient to demonstrate the calculator's math: a 55-year-old South Asian man with treated hypertension (that is, both diagnosed with and on medication for hypertension), diabetes, non-smoking status, total cholesterol 213 mg/dL, HDL cholesterol 45 mg/dL, systolic blood pressure 145 mmHg, and a normal waist-to-hip ratio (WHR). Compared to a normal WHR, an elevated WHR is defined as WHR ≥ 0.90 (male) or ≥ 0.85 (female), per WHO central adiposity thresholds. This profile is referred to below as the illustrative profile. It is deliberately distinct from the validation profile used in Table 1 immediately below, which reproduces the ACC/AHA guideline's own published worked example (no hypertension, no diabetes) and exists only to confirm the calculator's baseline PCE computation is correct.

Model A begins with the standard PCE, computed using the published White-cohort coefficients from the 2013 ACC/AHA Full Work Group Report, the only PCE coefficient set that exists, since no South Asian-specific set has been published.[3] The PCE equation itself takes no ethnicity or country-of-origin input; the calculator applies this same fixed coefficient table to every patient regardless of which country of origin is selected, and the country-of-origin ratio described below (C) is layered on afterward as a separate step. To confirm the coefficient table was implemented correctly, the calculator was run on the exact age, cholesterol, HDL, blood pressure, smoking, and diabetes values the ACC/AHA report itself used in its published worked example, a 55-year-old man and a 55-year-old woman with total cholesterol 213 mg/dL, HDL 50 mg/dL, untreated systolic blood pressure 120 mmHg, nonsmoker, no diabetes, and the output was compared against the report's own published result. Table 1 reports the comparison.

Profile	ACC/AHA report	Calculator output	Difference
Man, age 55 (report's example inputs above)	5.3%	5.38%	+0.08 pp
Woman, same inputs	2.1%	2.05%	−0.05 pp

Table 1. Validation of the PCE coefficient table against the ACC/AHA source report's own published worked example. The country-of-origin selector plays no role at this step. pp = percentage points.

The small residual differences arise because the source report rounds intermediate natural-log values to two decimal places before multiplying, while the calculator carries full floating-point precision throughout, the same approach used by other independent PCE reproductions (for example, the CVrisk R package), which produces a marginally more, not less, accurate result.

A South Asian recalibration multiplier is then applied to the PCE's output. Published comparisons between PCE-predicted and observed risk in South Asian cohorts show that the size of this correction depends on how high or low a patient's PCE-predicted risk already is, so a single flat multiplier cannot represent it. Two independent cohorts provide the data used to build a risk-dependent multiplier instead. Pursnani et al. (2022), a Kaiser Permanente Northern California cohort of 1,835 South Asian adults, found that in the cohort's lowest-risk stratum the PCE predicted a 1.8% ten-year ASCVD risk against an observed risk of 4.9%, meaning the true risk was 2.72 times the PCE's estimate.[7] Patel et al. (2021), a UK Biobank analysis of 8,124 South Asian participants, found that across the full cohort the PCE predicted a median 4.8% ten-year risk against an observed cumulative incidence of 6.8%, a 1.42-fold underestimate.[8]

Therefore, the correction factor should change dynamically based on the patient's baseline PCE risk rather than applying a constant adjustment. To achieve this, the multiplier is modeled by fitting a straight line between two anchoring points after converting both the PCE risk and the multiplier into logarithms, which is mathematically equivalent to fitting a power-law curve on the original scale. A constant additive correction would distort risk adjustments at the clinical extremes, such as inappropriately over-amplifying low risks or dampening high risks. Therefore, by electing to use a power-law relationship, the correction factor ensures that the adjustment acts as a proportional scaling factor. For a patient whose uncorrected PCE risk falls between 1.8% and 4.8%, the multiplier is read off this curve. For a patient whose uncorrected PCE risk falls outside that range, below 1.8% or above 4.8%, the multiplier is held flat at whichever anchor (2.72 or 1.42) is nearer, rather than extrapolated further along the fitted line. With only two calibration points, from two cohorts with different countries and case mixes, available, no evidence describes how the true correction behaves beyond the range these two studies actually observed.

Written formally: let R denote a patient's uncorrected PCE ten-year risk (as a percentage), and let M(R) denote the multiplier assigned at that risk level. Using the two anchor points above, R1 = 1.8% with M1 = 2.72, and R2 = 4.8% with M2 = 1.42, the multiplier is defined piecewise:

$$M(R) = M1, \quad \text{for } R \leq R1$$

$$M(R) = M2, \quad \text{for } R \geq R2$$

$$\ln M(R) = \ln M1 + [\ln M2 - \ln M1] \times [\ln R - \ln R1] / [\ln R2 - \ln R1], \quad \text{for } R1 < R < R2$$

The calculator's recalibrated risk estimate is then R × M(R) × C, where C is the country-of-origin ratio described next.

C is an optional country-of-origin ratio, derived from Patel et al.'s South Asian ethnic-subgroup hazard ratios against a White European reference (Bangladeshi 3.66, Pakistani 2.45, other South Asian 2.41, Indian 2.08), each divided by the same cohort's overall South Asian composite hazard ratio (2.03).[8] In other words, compared to a White European reference, Bangladeshi-origin participants carried roughly 3.66 times the risk, Pakistani-origin participants roughly 2.45 times, other South Asian participants roughly 2.41 times, Indian-origin participants roughly 2.08 times, and South Asians overall roughly 2.0 times. Dividing by the composite hazard ratio centers C at 1.00 for an unspecified South Asian origin and scales the recalibration multiplier proportionally up or down when a specific country is given. Table 2 reports both the source hazard ratios and the resulting ratio applied by the calculator.

Country of origin	**South Asian-vs-White European HR (Patel et al. 2021)**	**Country ratio applied (HR ÷ pooled composite HR of 2.03)**
Pooled / unspecified	2.03 (composite)	1.00
Bangladeshi	3.66	1.80
Pakistani	2.45	1.21
Other South Asian	2.41	1.19
Indian	2.08	1.02

Table 2. Derivation of Model A's country-of-origin refinement ratio from Patel et al. (2021).[8] HR = hazard ratio.

Applying the illustrative profile defined above (55-year-old unspecified South Asian man with on hypertension medication, with diabetes, non-smoking status, total cholesterol 213 mg/dL, HDL cholesterol 45 mg/dL, systolic blood pressure 145 mmHg, and a normal WHR), the PCE computes an uncorrected risk of R = 17.58%. Because 17.58% exceeds the upper calibration anchor of 4.8%, the multiplier is held flat at M = 1.42. With no country of origin specified, C = 1.00, and the recalibrated risk is 17.6 × 1.42 × 1.00 = 24.96%. If Bangladeshi origin is specified instead, C = 1.80 (Table 2), and the recalibrated risk becomes 17.6 × 1.42 × 1.80 = 44.93%, the figure reported in Section 3.1.

2.4 Model B: Stroke/ICH Subtype-Specific Relative Risk Score

Model B is a relative-risk score built specifically for ischemic stroke and ICH, rather than for composite atherosclerotic cardiovascular disease. Using regional data from the INTERSTROKE study for South Asia, the model estimates combined stroke risk by multiplying subtype-specific odds ratios. For example, an odds ratio of 3.61 for ischemic stroke means that individuals with hypertension have 3.61 times the odds of having an ischemic stroke compared to those without it. The other risk factors included are current smoking (1.32 ischemic / 1.35 intracerebral hemorrhage), diabetes (1.16 / 3.23), and high waist-to-hip ratio (1.47 / 2.80). Because stroke is a relatively rare condition, these case-control odds ratios are treated as hazard ratios [6]. Model B is presented as a conceptual extension, intended to demonstrate how subtype-specific risk structure may be incorporated when only summary-level data are available.

This combined relative risk is applied to a South Asian diaspora baseline incidence rate drawn from Khan et al.'s age- and sex-standardized Ontario and British Columbia data: 39.7 ischemic strokes and 5.0 ICH events per 100,000 person-years [10]. In practical terms, this means that among 100,000 South Asian adults at the reference age of 55 years with none of the four modeled risk factors, an average of 39.7 adults would be expected to have a new ischemic stroke, and 5.0 would be expected to have a new ICH, over one year of follow-up. These two figures serve as the starting point to which the relative-risk multiplier, age-scaling factor, and country-of-origin ratio are then applied.

Furthermore, age-scaling in Model B combines a standard epidemiological heuristic, which assumes a doubling of incidence approximately every decade beyond the reference age of 55, with a subtype-specific shape modifier derived from Vyas et al. (2021). That study examined a province-wide Ontario cohort of 243,843 South Asian immigrants and reported age-stratified, stroke-subtype-specific hazard ratios comparing immigrants to long-term residents across seven age bands.[11] Table 3 below reproduces the printed values of the aforementioned paper’s Figure 2 as a table for readability. The reference band used throughout this manuscript, ages 50 to 59 (containing this project's reference age of 55), has an adjusted HR of 0.78 for ischemic stroke and 1.04 for intracerebral hemorrhage, again taken directly from Figure 2's printed labels. The shape modifier at a given age is calculated as that age band's hazard ratio divided by the reference band's hazard ratio, so the modifier equals 1.00 exactly at age 55; Table 3 reports the resulting modifier, for both stroke subtypes, at every age band.

Age-band midpoint (years)	Immigrant-vs-resident HR, ischemic stroke	Immigrant-vs-resident HR, ICH	Shape modifier, ischemic (HR ÷ 0.78)	Shape modifier, ICH (HR ÷ 1.04)
24	0.84	1.01	1.08	0.97
35	0.68	1.06	0.87	1.02
45	0.69	0.98	0.89	0.94
55 (reference)	0.78	1.04	1.00	1.00
65	0.82	1.03	1.05	0.99
75	0.74	0.86	0.95	0.83
85	0.59	0.69	0.76	0.66

Table 3. Age-banded immigrant-vs-resident hazard ratios from Vyas et al. (2021), pooled across all immigrant source regions, and the resulting age-scaling shape modifier applied by the calculator at each age band.[11] HR = hazard ratio. Vyas et al. (2021) report these hazard ratios in Figure 2. Each panel gives an unadjusted and an adjusted HR with a 95% CI per age band; only the adjusted point estimate is used here and in the calculator. Confidence intervals are omitted from Table 3 accordingly.

This age-banded curve is not South Asian-specific in isolation, it is pooled across all immigrant source regions in Vyas et al.'s cohort, but it is real, measured, subtype-specific age-varying data from the same population type (diaspora, Western health system) underlying the rest of this project, and represents a substantive improvement over an ethnicity-blind generic heuristic. Because this component is a cross-population proxy rather than a South Asian-specific measurement, it is named explicitly as a priority gap for future data collection in Section 4.5, alongside the other three gaps summarized in the Conclusion.

Putting these components together: for a given age, sex, and set of selected risk factors, adjusted annual incidence rate = baseline incidence (per subtype, Table 3 above) × combined relative risk × age-scaling factor × country-of-origin ratio. This annual rate is converted to a 10-year cumulative risk assuming a constant annual hazard: 10-year risk = $1 - e^{(-\text{annual rate} \times 10)}$.

Using the same illustrative profile (Section 2.3): hypertension and diabetes present, nonsmoker, normal waist-to-hip ratio, no country of origin specified.

RR(ischemic) = 3.61 (hypertension) × 1.16 (diabetes) = 4.19

RR(ICH) = 4.37 (hypertension) × 3.23 (diabetes) = 14.12

Age-scaling factor at age 55 (the reference age) = 1.00 for both subtypes

Adjusted annual rate (ischemic) = 39.7 × 4.19 × 1.00 × 1.00 ÷ 100,000 = 0.001663 per year

10-year risk (ischemic) = $1 - e^{(-0.001663 \times 10)} \approx 1.6\%$

Adjusted annual rate (ICH) = 5.0 × 14.12 × 1.00 × 1.00 ÷ 100,000 = 0.000706 per year

10-year risk (ICH) = $1 - e^{(-0.000706 \times 10)} \approx 0.7\%$

These figures match the illustrative outputs reported in Section 3.1.

It is worth noting that the multiplicative combination of INTERSTROKE odds ratios is treated as a known simplification: it assumes each risk factor's effect is independent of the others, which is unlikely to hold when three or more major risk factors co-occur. Rather than introduce an unvalidated additional correction for this, Model B flags any output where three or more major risk factors are present simultaneously as lower-confidence, making the limitation visible rather than silently absorbing it into a point estimate.

2.5 Validation Approach

No individual-level South Asian stroke or ICH cohort dataset was available for this project. Therefore two complementary checks were used instead. The first is a numerical robustness check, confirming that outputs remain within a plausible 0-100% range across a wide span of ages, sexes, and risk-factor combinations, and that the calibration curve reproduces its two defining anchor points exactly. The second is a cross-check against independent cohorts not used in model construction, most notably SABRE, which provides an external check on whether standard risk tools under-serve South Asian populations, broken out by sex, a dimension absent from Model A's own calibration anchors.[9] What these checks can and cannot establish is discussed in Sections 3.2 and 5.

3. Results

3.1 Calculator Overview and Illustrative Outputs

The tool is implemented as a self-contained, browser-based calculator (Supplementary File 1) requiring no installation, alongside a reference Python implementation (Supplementary File 3) used for development and validation; the two implementations were cross-checked and produce identical output on matched test cases. Across tested profiles, recalibration consistently increased estimated risk in low- and intermediate-risk strata, consistent with prior cohort observations. For the illustrative profile defined in Section 2.3, a 55-year-old South Asian man with treated hypertension and diabetes, non-smoker, pooled country-of-origin, and normal WHR, Model A recalibrates the PCE's uncorrected 17.58% ten-year risk estimate to 24.96%, and rises further to

44.93% if the Bangladeshi-origin country refinement is applied, an increase the tool flags explicitly as a lower-confidence, compounded estimate rather than presenting without qualification. Model B, for the same profile, returns a 4.19-fold relative risk for ischemic stroke and a 14.12-fold relative risk for ICH relative to a South Asian adult with none of the four modeled risk factors, translating to approximate ten-year absolute risks of 1.6% (ischemic) and 0.7% (ICH). Model B's age-scaling component draws on hazard ratios pooled across all immigrant groups rather than measured specifically in South Asian immigrants (Section 2.4, Table 3); this is discussed as a priority gap for future work in Section 4.5. Because Model B is not calibrated against individual-level outcome data, these estimates can be interpreted as illustrative of the role of certain risk factors in stroke.

3.2 Validation Findings

The calculator is internally consistent and numerically robust: no combination of inputs tested produced an out-of-range, negative, or undefined output (unless patient values were input as 0), and the recalibration curve's two defining anchor points are reproduced exactly. An earlier version of Model A used a single flat South Asian multiplier; stress-testing against the Pursnani and Patel cohorts revealed that this flat multiplier undershot the observed low-risk-stratum result by approximately 47%, since a single multiplier cannot simultaneously match both a 2.72-fold and a 1.42-fold correction at different points on the risk spectrum. Replacing it with the two-point calibration curve described in Section 2.3 resolved this by construction. Matching two calibration points confirms only that the fix was implemented correctly; it is not independent evidence that the curve generalizes to unobserved regions of the risk spectrum. Externally, the model behavior was qualitatively compared against independent cohort findings not used in model construction to assess whether the direction and magnitude of risk differences were consistent with reported epidemiologic patterns, discussed in 3.3.

3.3 Corroborating Evidence Across Independent Cohorts

Several independent lines of evidence, not used to derive any single coefficient in the tool, converge on its core premise that standard risk tools under-serve South Asian populations in some respect. SABRE found that both QRISK2 and Framingham substantially under-predict composite cardiovascular risk in South Asian women (predicted-to-observed ratios of 0.52 and 0.43, respectively) despite showing their best discrimination in that same subgroup (area under the receiver operating characteristic curve up to 0.77), and that conventional high-risk cutoffs would miss 57-67% of South Asian women who went on to have a cardiovascular event.[9] MASALA independently found that high PCE-predicted cardiovascular risk was associated with the presence of coronary artery calcium, a direct anatomical marker of atherosclerosis, in a US South Asian cohort (odds ratio 1.81 in men, using the 10-year risk metric; 1.97 in men and 3.14 in women, using the lifetime-risk metric), evidence that elevated risk estimates in this population track a measurable disease marker and not only self-reported outcomes.[16] Razieh et al. (2022), an independent UK Biobank sub-cohort not used to build either model, found that South Asian ethnicity carried elevated cardiovascular risk (hazard ratio 1.24, 99% CI 1.03-1.48) that survived adjustment for a wide range of sociodemographic, lifestyle, environmental, and clinical covariates.[12] Patel et al. separately found that hypertension, diabetes, and central adiposity each explain a significantly larger share of cardiovascular disease variance in South Asians than in Europeans, supporting Model B's decision to weight these factors using South Asia-specific rather than generic coefficients.[8] INTERHEART independently found that its nine conventional risk factors account for 89.4% of myocardial infarction risk in its South Asia region cohort, closely matching the 90.8% of all-stroke risk that INTERSTROKE separately attributes to its ten risk factors in the same region, cross-validating, across two unrelated multinational studies, that standard modifiable risk factors explain a comparably large share of both cardiac and cerebrovascular risk in South Asians.[4,6]

4. Discussion

Rather than presenting a validated clinical prediction model, this work introduces an evidence-synthesis-based prototype that illustrates how recalibration strategies can be constructed from currently available data. Four findings from this synthesis are highlighted below from the development of the calculator and analysis of data:

4.1 Unresolved Direction of PCE Miscalibration in South Asians

The clearest and most consequential tension in the evidence base concerns the very premise of Model A's recalibration. Pursnani et al. and Patel et al., the two cohorts underlying Model A's calibration curve, both found that the PCE underestimates observed South Asian risk.[7,8] Rodriguez et al. (2019), a Northern California electronic health record cohort of 231,622 adults that disaggregates Asian Indian participants specifically (n=13,815) rather than pooling them into a broader Asian or South Asian category, found the opposite: the PCE overestimated risk in Asian Indians by approximately 30% (predicted 1.2% versus observed 0.9%, predicted-to-observed ratio 1.3, $p<0.001$).[13] Therefore, this study is a direct disagreement about which way the correction should point.

A closer reading of Pursnani et al.'s own data offers a partial, though not complete, reconciliation. Their headline low-risk-stratum result (a 2.72-fold underestimate) pools patients across three statin-exposure categories, including those who initiated statin therapy during follow-up, plausibly because a clinician judged them at elevated risk. The subgroup of patients never prescribed a statin, the closest available match to Rodriguez et al.'s not-yet-treated cohort, showed a substantially weaker and statistically non-significant gap (predicted 1.4% versus observed 2.1%, ratio 0.67, $p=0.12$).[7] This raises the possibility that some portion of the headline underestimation figure reflects confounding by treatment indication rather than a pure calibration failure. It does not, however, fully explain the discrepancy with Rodriguez et al., whose cohort differs in geography, timeframe, risk window (5-year versus 10-year), and ethnic granularity (Asian Indian specifically versus a broader South Asian definition that includes

Pakistani, Bangladeshi, Sri Lankan, and Nepali individuals). Both directions are presented in this manuscript rather than adopting one, and Model A's multiplier was not rebuilt around this single additional data point. This is regarded as an open, well-defined empirical question that future studies may address to reconcile.

4.2 Family History Coefficient

Family history is a well-established stroke risk factor in general populations, and was among the variables this project originally set out to include. However, no South Asian-specific family history hazard ratio or odds ratio was located anywhere in this synthesis, despite a targeted search across every major source examined. Gunarathne et al. (2009) and Khan et al. (2013) each report family history prevalence in South Asian stroke patients without an associated effect estimate.[14,15] Vyas et al. (2021) explicitly states that family history was among the variables their administrative dataset did not capture.[11] Therefore, while family history is established as a risk factor, the absence of data specifically for South Asians offers yet another direction that must be explored to refine the calculator. In this study, family history has accordingly been omitted from both models rather than substituted with a generic, non-South-Asian-specific coefficient.

4.3 Sex as a Modifier

SABRE's finding that standard risk tools calibrate far worse in South Asian women than men (Section 3.3) raised the possibility that Model A's recalibration multiplier should differ by sex. A sex-stratified predicted-versus-observed PCE comparison in South Asians was searched for specifically, examining Pursnani et al., Patel et al., and Rodriguez et al. in full text, including supplementary tables.[7,8,13] None reports one. Patel et al. does test whether the underlying South Asian-versus-European cardiovascular hazard ratio itself differs by sex and finds no significant interaction (women hazard ratio 1.92, 95% CI 1.62-2.28; men 2.06, 95% CI 1.87-2.28; p-heterogeneity=0.12), a related but distinct question from SABRE's finding, which concerned the calibration performance of specific risk scores (QRISK2, Framingham) rather than the raw ethnicity-associated hazard ratio.[8,9] This question is considered investigated rather than resolved: the evidence is genuinely mixed, and no unsupported sex-specific correction has been built into Model A on the strength of a single, narrower-scope finding.

4.4 The Age-Scaling Curve Is Pooled Across Immigrant Groups, Not South Asian-Specific

Model B's age-scaling shape modifier (Section 2.4, Table 3) is drawn from Vyas et al. (2021), a cohort of 243,843 immigrants to Ontario across all source regions, not a South Asian-specific age-stratified stroke cohort.[11] It was used here because it is, to this project's knowledge, the only available source providing genuinely age-banded, stroke-subtype-specific hazard ratios for a diaspora population in a Western health system, a substantive improvement over an ethnicity-blind generic doubling-per-decade heuristic. It remains, however, a pooled-immigrant approximation rather than a direct South Asian-specific measurement, and the true shape of the age-risk curve for South Asian stroke specifically may differ from the pooled curve used here in ways this tool cannot currently detect. A South Asian-specific, age-stratified, subtype-specific stroke incidence dataset would allow this component of Model B to be replaced with a directly measured curve, and is named here as a fourth priority for future data collection, alongside the three described in the Conclusion.

4.5 Clinical and Public Health Implications

This tool is a research prototype built entirely from published, cohort-level summary statistics, not from patient-level data, and has not been tested against a held-out clinical dataset because none was available for this project. It is not intended, and should not be used, as a stand-alone clinical decision-making instrument. Its value is threefold. First, it is, to this project's knowledge, the first synthesis to assemble the scattered South Asian stroke and ICH literature into a single, internally consistent, subtype-specific framework, replacing an implicit assumption (that generic tools are approximately correct) with an explicit, revisable one. Second, by flagging low-confidence outputs and stating its calibration anchors openly, it demonstrates a pattern for how ethnicity-specific risk tools might communicate uncertainty. Third, it gives the unresolved tensions described in this Section a concrete, falsifiable form: specific curves, built from specific data, that the next South Asian cohort study with individual-level PCE performance data, ideally sex-stratified, could directly test and either confirm or overturn.

5. Limitations

This work has several limitations beyond those already discussed. No patient-level validation was performed; every coefficient derives from published cohort-level summary statistics, and the tool's absolute risk outputs should be read as illustrative rather than clinically calibrated. Model A's calibration curve is fit through exactly two points from two cohorts with different countries, healthcare systems, and case mixes; it is a defensible improvement over a flat multiplier but not a validated dose-response curve. Model B's age-scaling shape modifier, though genuinely subtype-specific and age-stratified, is drawn from a pooled-immigrant cohort rather than a South Asian-specific one (Section 4.5). Model B's multiplicative combination of INTERSTROKE odds ratios assumes risk-factor independence, which likely overstates risk when three or more major risk factors co-occur; this is flagged rather than corrected, since no South Asian-specific multivariable (jointly estimated) stroke risk model was located to replace it. INTERSTROKE's odds ratios are case-control estimates, treated here as approximating hazard ratios under the rare-disease assumption, standard practice for stroke research but an approximation nonetheless. Finally, this tool is scoped to South Asian-origin adults in Western diaspora settings and is not intended for use in South-Asia-resident populations, whose risk profile, per Khan et al., may differ substantially from the diaspora cohorts this tool is built from.[10]

6. Conclusion

This study presents an open-source, dual-model prototype illustrating how incorporation of South Asian-specific data alters cardiovascular and stroke risk estimation, while explicitly identifying structural limitations in the current evidence base. Honest documentation of these gaps, a missing family history coefficient, an unresolved disagreement over the direction of PCE miscalibration, an unresolved disagreement over country-of-origin risk gradients, an unanswered question about sex as a calibration modifier, and Model B's reliance on a pooled-immigrant rather than South Asian-specific age-stratified stroke risk curve, is regarded as no less substantive a contribution than the calculator itself. Each gap described here is specific enough to be directly tested by future work; therefore, this synthesis is intended to accelerate, not substitute for, the patient-level validation the field still needs.

Data and Code Availability

The calculator is provided as Supplementary File 1. a self-contained HTML/JavaScript application that runs entirely in a web browser with no installation or internet connection required after download. A reference Python implementation used for development and validation is provided as Supplementary File 3. The complete evidence-tracking document underlying this manuscript, including every source's sample size, effect estimate, confidence interval, and verification status, is provided as Supplementary File 2.

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